NATHAZ AI LAB / TECHNICAL REPORT · EXPLORATORY TRIAL

R23.2b High-wind forecasting and active control for buildings

Group AI Weather Models

Status Closed (negative result)

Period 2026-10-14 14:35 — 2026-10-24 22:02

Stage conclusion

Project report

A2 Building high-wind forecasting and active control

Objective

Evaluate three control layers across time scales: hour-to-day mode switching, second-to-minute feedforward, and real-time feedback. Determine which layer AI weather forecasts can supply Control covers active mass damper systems and active aerodynamic shape control, with wind as the common input.

Current status

Node opened on 10-14. Step 1 at 14:38: Grok quick search returned at 14:39 (abstract-level only). Step 2 at 14:5x: Deep Research (`--gemini`) launched, reusing the feedforward conclusion of the A3 Grok report without a

new search (whose Section 6 numbers were later rejected by the A3 review panel as unsourced and replaced with verified A3 citations). Original plan: after receiving the report and reading the A3 Grok report, write the second Deep Research (`--gemini`) brief and send conclusions for review.

Deep Research returned at 15:0x. The lead verified the sources and sent the package to a 3-member review panel at 15:1x.

After the review (reviewer E with reservations, reviewer A not supportive, reviewer D with reservations), the ruling cancelled the original minimal trial (AI forecast → Tokyo Polytechnic University (TPU) wind tunnel data → threshold exceedance). A new minimal trial was assigned to the ideas agent (queued): local Silsoe lattice Boltzmann method (LBM) lead-time τ prediction with stroke-limited active mass damper preview feedforward on workstation CPU (0 compute units, CU).

Three-layer table (finalized after review)

Time scale	Control action	Required lead time and accuracy	Precedents (source and page)	Can AI forecasts supply it?
Hours–days	Slow actions only: mass recentering and mechanical locking, hydraulic/backup power preparation with manual review, slow mechanical shape changes or tuned liquid damper (TLD) filling/draining, whole-storm stroke budgeting	Execution time and latest decision point not reported in literature; "requires hours" is an assumption	No written operating procedures for forecast-triggered switching found in three independent searches (not claimed to be nonexistent); deployed systems trigger in real time via acceleration/stroke thresholds (Ping An Finance Center, PAFC: Zhou & Li 2022 Eng Struct 269:114840 abstract, damping ratio 0.6%→2–7%, no threshold given)	Yes, with small marginal gain: official typhoon/hurricane warnings already cover slow actions, while global AI models lack hurricane intensity skill (DeMaria 2025 card, `event_step0`, line 14) → **On hold**

Time scale	Control action	Required lead time and accuracy	Precedents (source and page)	Can AI forecasts supply it?
Seconds–minutes	Near-field preview feedforward / preview model predictive control (MPC); gain reduction and semi-active gear shifting (milliseconds–seconds, real-time sensing suffices)	Preview seconds: no verified source (A3 Grok §6 numbers discarded after A3 review); wind-turbine lidar feedforward works in stationary turbulence: Guo 2023 WES 8:149 §5.2 tower-base fore-aft damage equivalent load (DEL) up to −16.7%, Bossanyi 2014, Schlipf 2018 (verified in A3)	No engineering demonstration; research/concept level: Abdelaziz & Hobeck 2022 SPIE 12046 (10.1117/12.2630805, surrounding building pressures + gated recurrent unit, GRU, window + aerodynamic corner flaps, 1:400 wind tunnel, root-mean-square, RMS, rotational acceleration up to −75%, title verified, content via reviewer D); US5592791A (1997 patent, corner boom upstream anemometer feedforward, original verified by lead); Klein 1972, Soong & Skinner 1981, Abdel-Rohman 1983 (via Korkmaz 2011 p. 2130); theory in Marzbanrad 2004 Eng Struct pp. 1463–1471	No: requires on-site near-field wind sensing; **narrow focus of this node**; local LBM τ=0 correlation drops from 0.67 at 7 m to 0.35 at 27 m (local data)
Real-time	Acceleration / displacement feedback	None	Standard practice	Not needed

Preview statistics result: Near-field velocity provides second-scale preview for windward pressure coefficient Cp (correlation half-life 2.6–6.5 s, local data). Control round 2 by the ideas agent (dispatched 10-14 16:1x) first checks the control model (unsaturated perfect preview must cut acceleration root-mean-square, RMS, by ≥30%) before measuring the fraction of perfect-preview gain captured by predicted preview.

One-sentence conclusion (decided 10-14): On hold. Exploratory; not peer reviewed. AI weather forecasts serve only hour-scale slow actions with minimal gain. Second-scale near-field preview offers large gains in principle (68–80% acceleration reduction under perfect preview), but sparse upstream probes recover only 0–17% of that gain because inflow prediction skill is the bottleneck. Reasons, missing conditions and restart conditions are documented separately.

(Earlier provisional note) AI weather-forecast-driven building control offers low value, so we place this layer On hold. Second-scale near-field preview feedforward is worth continuing: first quantify usable lead time and stroke-limited damper gain on local data, then decide whether to promote it or merge it into sub-project A3's downburst preview trial.

Decisions made

- 2026-10-14 literature ruling and scope narrowing.

Trials and results

Date	Action taken	One-sentence result
10-14	Grok quick search	Found almost no building/bridge precedents driven by upstream inflow preview or numerical weather prediction; existing "prediction" is mostly model predictive control (MPC) state extrapolation or disturbance estimation, while real active tuned mass dampers (ATMDs) trigger via acceleration thresholds (unverified fragments; 2004 DOI actually points to Xie 2004 interference effects; Taipei 101 TAO DOI not found; "0.75–1.5 s preview" was Grok's own calculation from this line's 30 m data, not literature evidence).
10-14	Minimal trial round 2 version 3 (`ideas`, workstation CPU 20.5 s, 0 compute units)	With corrected feedforward, perfect 8 s preview cuts acceleration root-mean-square (RMS) by 68% (near spectral peak) and 80% (off-resonance), whereas velocity-probe predicted preview captures only 0–17% of the perfect gain, passing the gate in just 4 of 360 combinations (near spectral peak, tight stroke, 10.3–12.6% reduction), so discrimination is weak; best off-resonance 0.8%, so inflow prediction skill is the bottleneck.
10-14	Minimal trial version 2 (`ideas`, workstation CPU 14.5 s, 0 compute units)	Preview statistics from a single local realization show windward 11-point pressure coefficient (Cp) correlation half-life lead time rising from 2.6 s (7 m standoff) to 6.5 s (27 m), or 2.7–6.6 s after notch filtering; all predicted-preview control combinations failed the gate, and **even perfect 8 s preview cut response by only 0.4–0.9%** and even the loose stroke tier hit its limit; the control model lacked discriminating power, so no control conclusion holds and round 2 was dispatched.

Date	Action taken	One-sentence result
10-14	Deep Research (`--gemini`)	Confirmed real inflow preview for buildings and bridges is largely absent, located Marzbanrad 2004, compiled an unverified operating table (Citicorp, Kyobashi Seiwa, Yokohama Landmark, Canton Tower, Ping An Finance Center), and found no forecast-switched operating rules (discarded its 30 m calculation, vehicle/jetway cases, and uncited pre-typhoon warm-up claim).
10-14	Lead check of original text Abdelaziz 2021	Verified (PDF p. 1 abstract): an artificial neural network (ANN) schedules flap angles from current wind conditions in steady two-dimensional fluid-structure interaction (2D FSI), cutting amplitude by up to 94% in preliminary results; this represents wind-condition scheduling rather than inflow preview.

Parameter source table

Parameter	Value	Source or "assumption"	Places checked
Ping An Finance Center (PAFC) ATMD trigger/lock threshold	Not found in public literature	Grok fragment 0.01 m/s² / 2.2 m not adopted	Grok, Deep Research (DR), Crossref/OpenAlex abstracts (unavailable)
Citicorp tuned mass damper (TMD) activation threshold	0.003 g for two cycles	Deep Research paraphrase (vibrationdata/buffalo webpage), original unverified	`pages` no hit
Kyobashi Seiwa active mass damper (AMD) stroke/force	±25 cm / 1 tf	Deep Research paraphrase, unverified	—
Minimal trial Tokyo Polytechnic University (TPU) scale/α	1/400, α=1/4, 1/6	TPU official case page (verified 10-13)	

Open items and next steps

- The correlation–τ–distance table was shared with the digital-twin wind tunnel root lead, who archived it without objection. With sub-project A2 On hold, they will no longer extract inflow dynamic pressure at the building location in the empty domain; on restart it can be requested from the digital-twin wind tunnel lead at any time.

- Hierarchy constraint (root lead 10-14): This node sits at level 2 and will not launch a sub-project at level 3; subdivisions use task briefs without re-delegation by tool bots.
- Workflow sequence: Receive Grok report → read A3 Grok report → draft Deep Research brief (`--gemini`) → run 3-reviewer panel → select GPU minimal trial candidate.
- Local data reference: Silsoe LBM inflow-to-windward Cp correlation drops monotonically with standoff distance (0.67 at 7 m → 0.35 at 27 m) (local data, not literature).

Literature basis

- Knowledge base (10-14 search): No cards on forecast- or preview-driven structural control. Local full texts (`pages`) provide leads awaiting cards: Abdelaziz et al. 2021 JWEIA 212:104601 smart facade control (PDF p. 1); Korkmaz 2011 active structural control review (PDF p. 18 = journal p. 2130, citing Abdel-Rohman 1983 aerodynamic appendages and Nissen 2004 bridge active aerodynamic stabilization); Kareem 1978 (PDF p. 55 = journal p. 32). Cards still to be created and checked.

PI notes

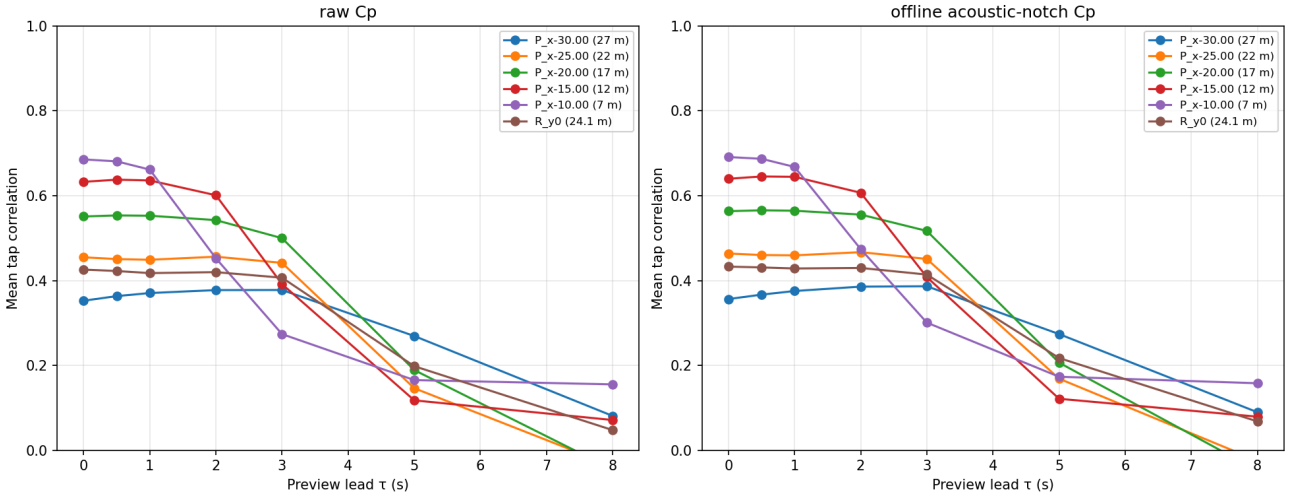
Merged into C39.8 (10-29, forecast-driven + preview control; this page remains On hold)

- **Ownership**: The scientific question of R23.2b merges into **C39.8** as "forecast-driven + preview control"; papers belong solely to C39.8. Sub-project R23.2b **remains On hold per the PI's 10-21 decision**, with no compute resumed and no gate changes; C39.8 remains plan-only until its launch plan is approved.
- **Revised one-sentence question**: How much net gain can active or semi-active control of buildings and bridges (stroke-limited AMDs and bandwidth-limited shape actuators) achieve at a given preview lead time and preview error? Map the "preview lead time × preview error → net gain" curve, give the minimum usable preview quality, and test whether slow actuators can use preview. Hour-scale AI weather forecasts serve only the mode-switching layer and will not be restarted separately.
- **Inputs handed from R23.2b to C39.8** (all local data from a single Silsoe LBM realization, no literature basis): 1. Perfect 8 s preview with stroke-limited active mass damper (AMD) preview feedforward cuts acceleration RMS by 68% (near spectral peak) and 80% (off-resonance). 2. Upstream velocity-probe predicted preview captures only 0–17% of perfect-preview gain; 4 of 360 combinations pass the gate (near spectral peak, tight stroke, 10.3–12.6% reduction), while the best off-resonance reduction is 0.8%. 3. Windward 11-point Cp correlation half-life lead time rises from 2.6 s (7 m standoff) to 6.5 s (27 m), or 2.7–6.6 s after notch filtering; τ=0 correlation drops from 0.67 at 7 m to 0.35 at 27 m. Note: this marks the correlation half-life point, not the zero-gain point (six-reviewer panel, reviewer A). 4. Hours–days layer: official warnings already cover slow actions, leaving little marginal gain.
- **Recommended pre-registered stop rule for C39.8** (six-reviewer panel, reviewers H and B): Declare Closed (negative result) if predicted-preview gain falls below 10% of perfect-preview gain; report only

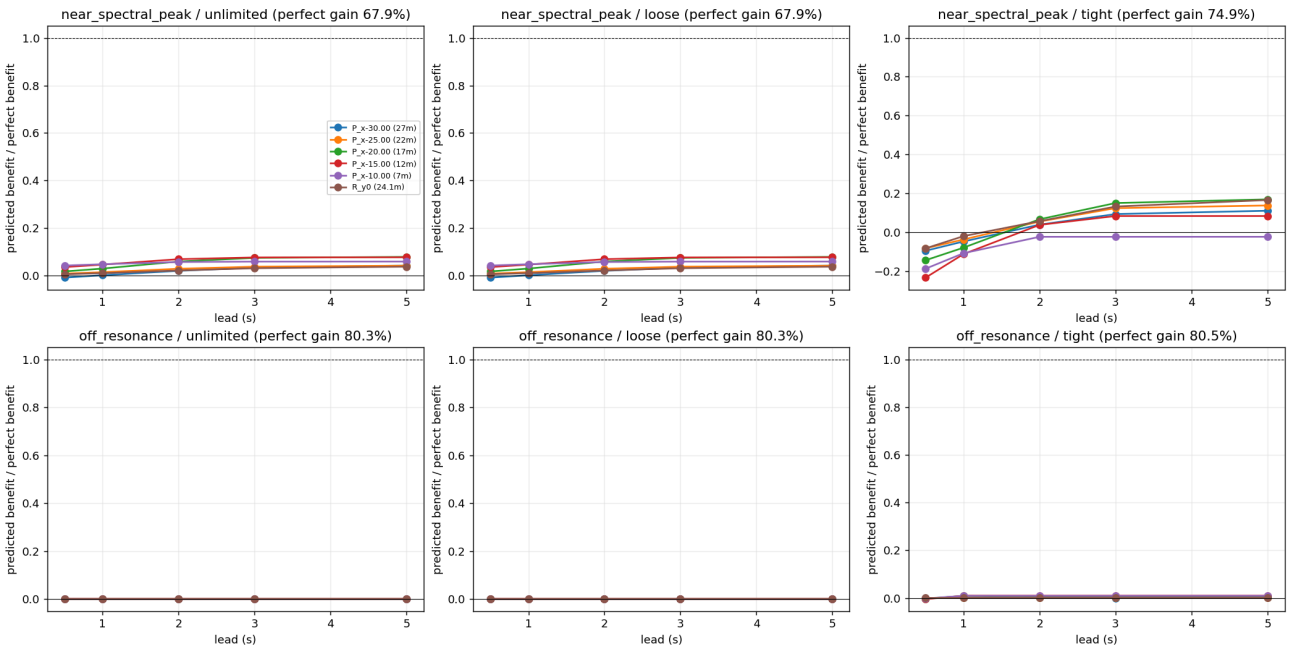
the surface and threshold without repeating the chapter 5 actor–critic demonstration from her dissertation.

- Summary handed to C39.8: (in English, group C rules).
- On 11-06, at the request of C39.8 (launched 11-06), delivered the frozen input manifest: SHA-256 hashes of source CSV and A2 files, coordinate/component/time-axis/window/Cp/q definitions, code definitions for gain ratio and the 10% gate, and known limitations; τ=0 correlation follows A2's 0.685/0.353 (9 PC, test window 522–634 s), whereas 0.67/0.35 in the 10-29 summary used version 4b (12 PC, 522–642 s). No compute run; R23.2b remains On hold.

Key figures



preview



control